

TOPICS FOR TODAY:

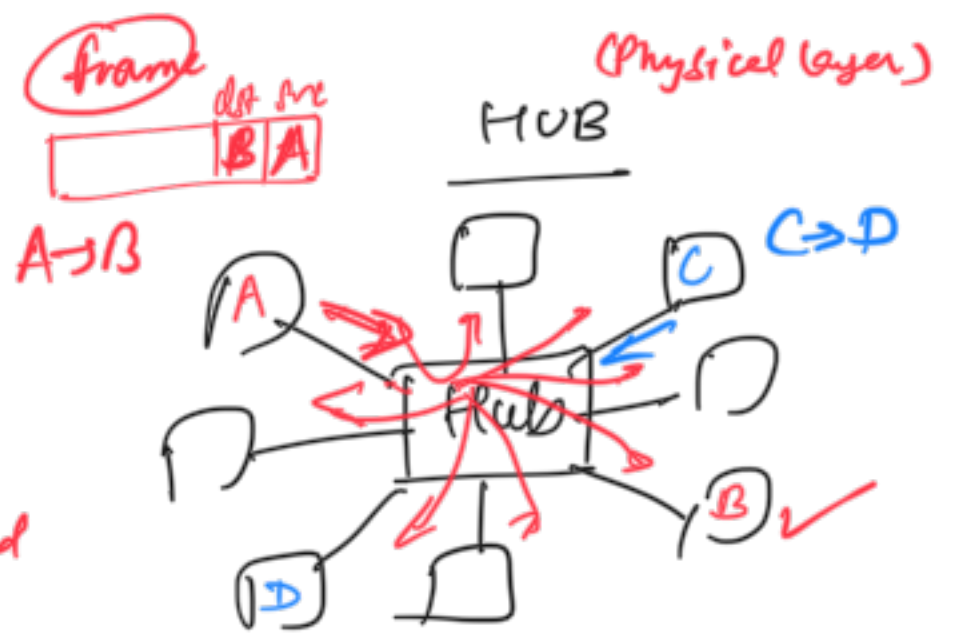
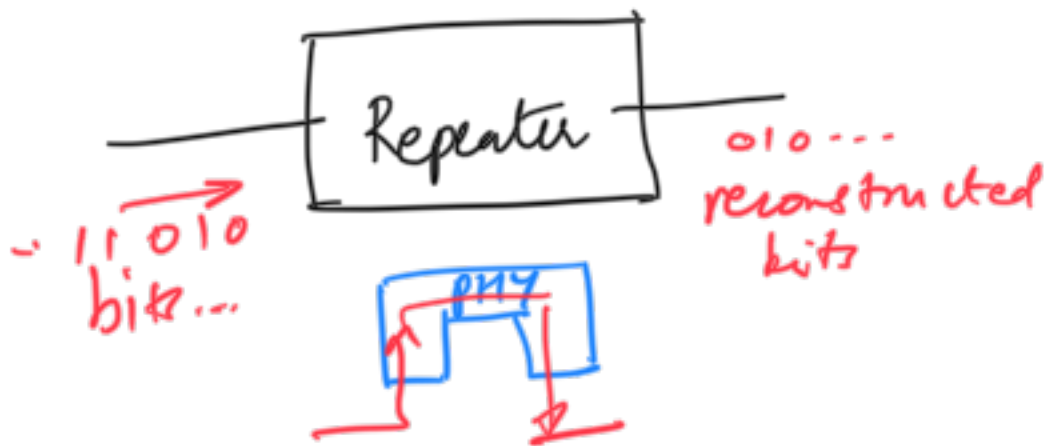
- Repeaters, Hubs, switches, routers

- ↳ Concept
- ↳ Collision domain
- ↳ Broadcast domain

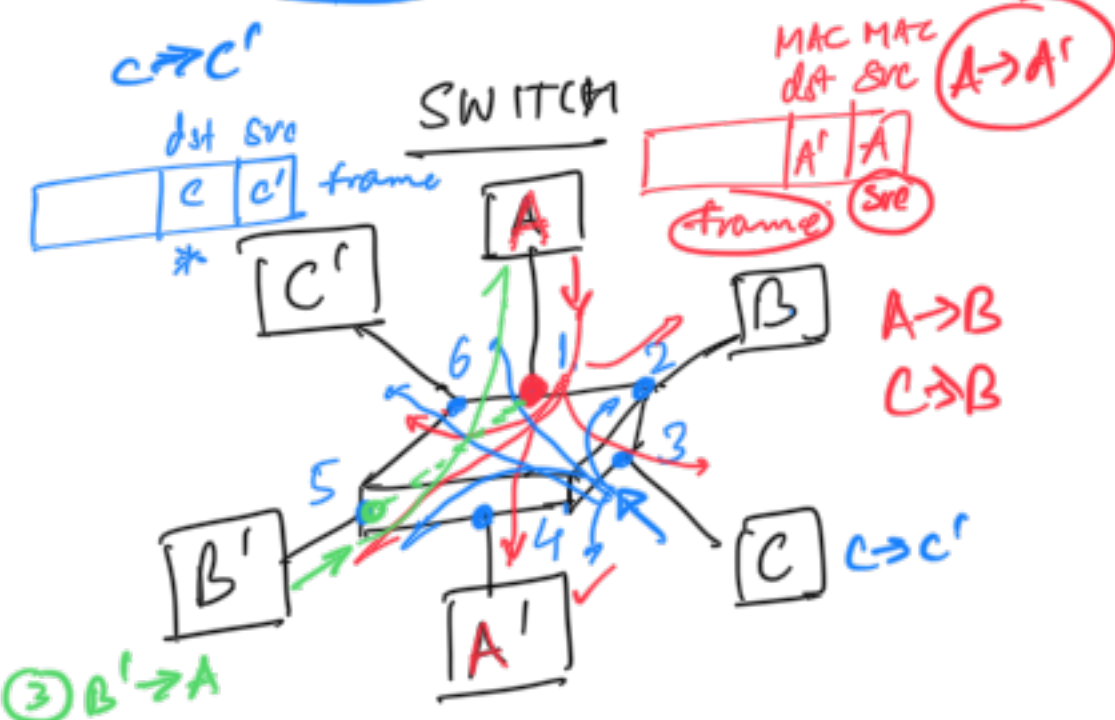
- Switch: Self learning (algorithm)
- Address resolution protocol (ARP)
- Virtual LANs (VLANs)
- Content Delivery Network (CDN)

REPEATER

physical



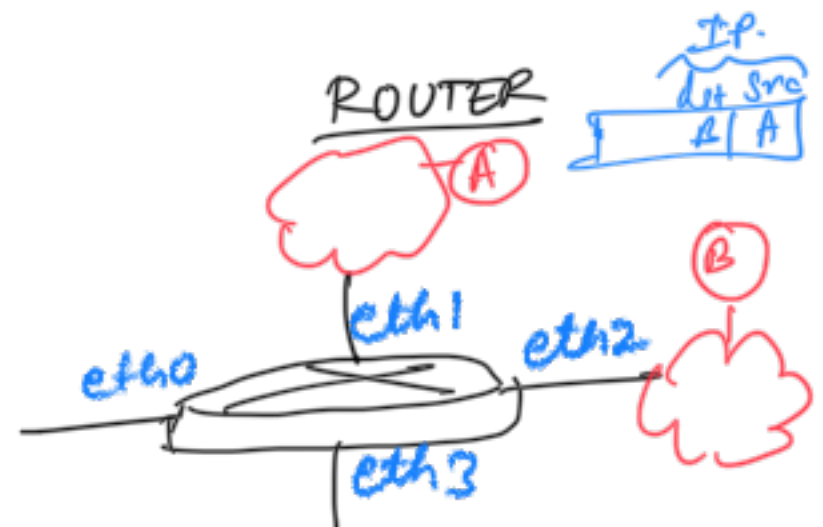
Layer 2 → Data link layer



Forwarding table

MAC addr	Interface	TTL
A	①	500
C	3	500
B'	5	500

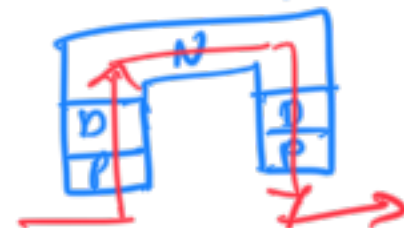
** in sec*



Routing table

Network addr mask	Next hop	Cost

Longest Prefix match (CPM)

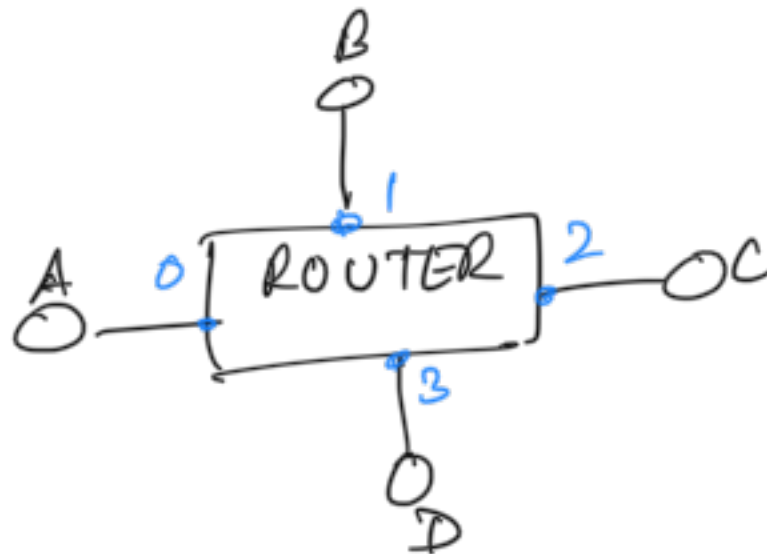
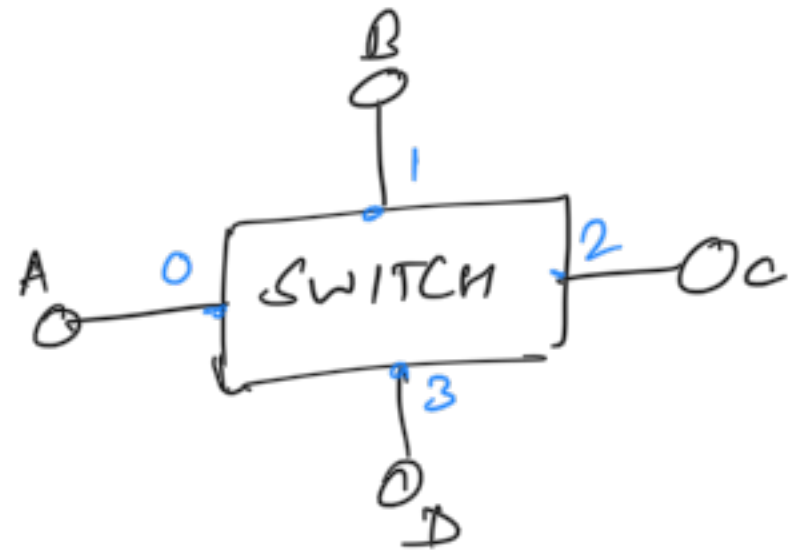
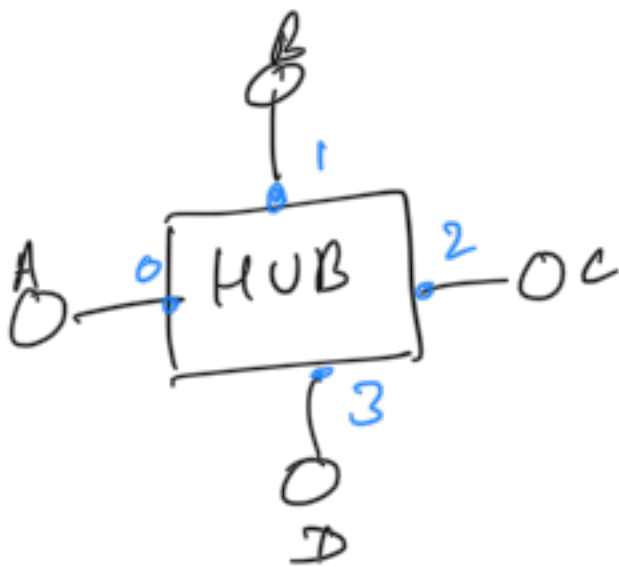


* Collision domain: area where a collision can occur

HUB	Single coll domain —
SWITCH	# coll domain = # ports —
ROUTER	# coll domain = # ports —

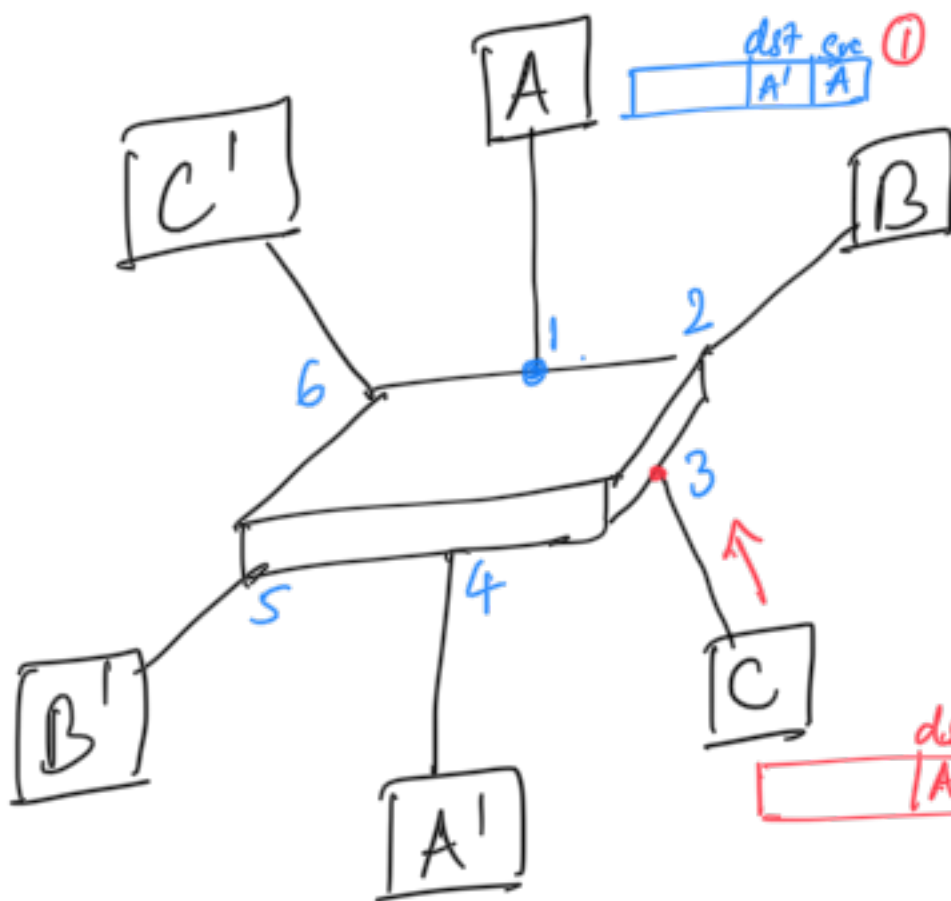
* Broadcast domain: area where a broadcast can reach

HUB	Single
SWITCH	Single
ROUTER	# broad domains = # ports.



* Switch learning algorithm (Plug-n-play devices)

ALGORITHM :



Initially the forwarding table is EMPTY
A new frame arrives at the switch

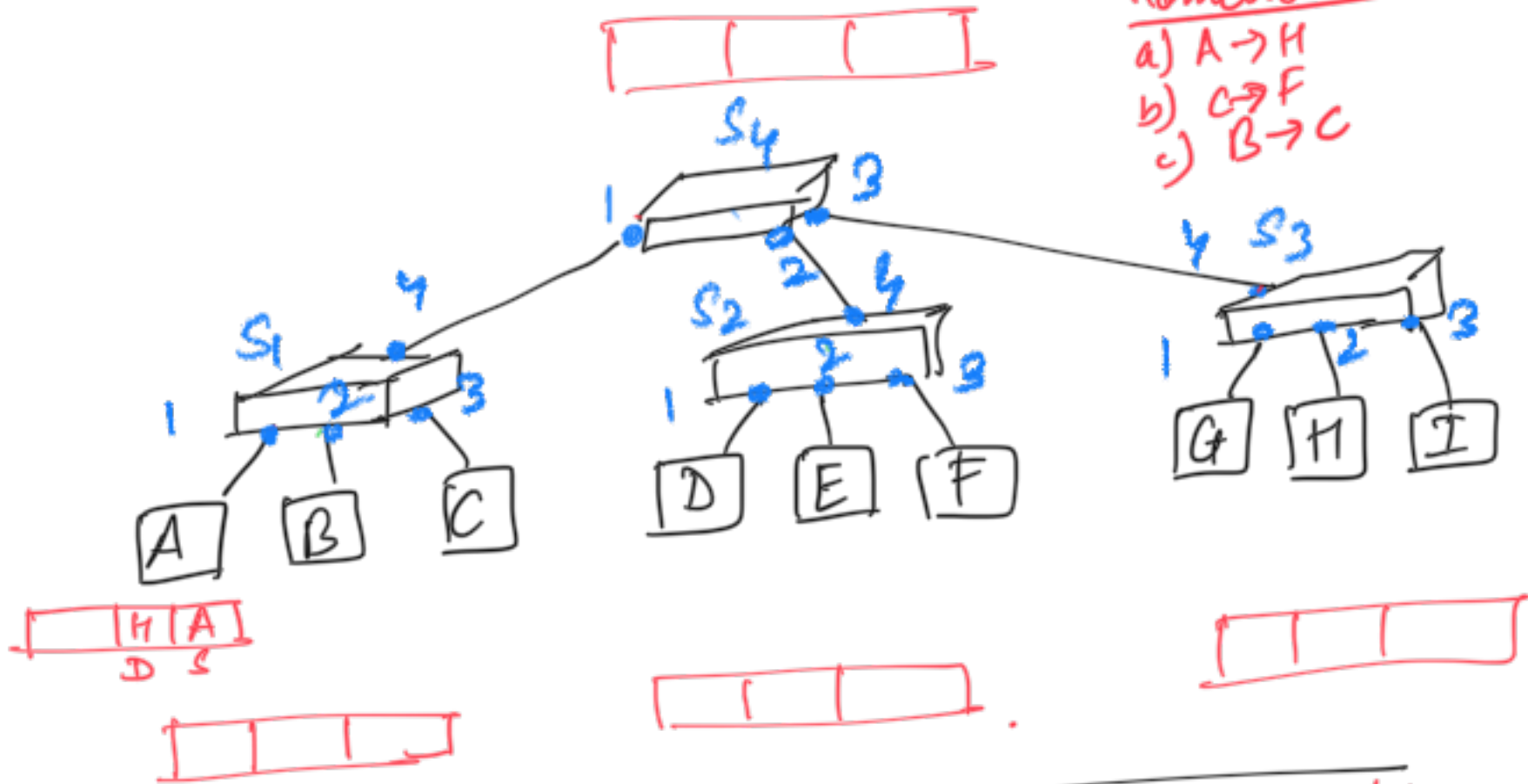
- ① Identify src addr, src interface
 - learn & make an entry into Forwarding table
- ② Identify the dst addr
 - HIT: Forward to up int
 - MISS: broadcast
- ③ Lookup for dst into forwarding table
 - (a) not found $\Rightarrow$ flood
 - (b) found $\Rightarrow$ forward
 - (c) if the destination interface/segment is same as the src interface $\Rightarrow$ discard the frame

Forwarding table		
MAC addr	Interface	TTL

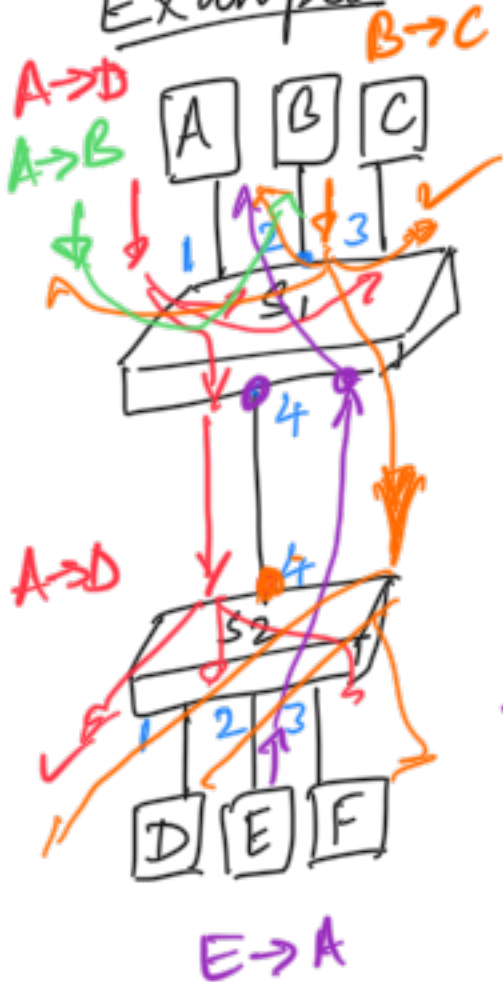
* Switch learning example (Hierarchical switches)

Homework:

- A → H
- C → F
- B → C



Example



Forwarding table: S1

MAC	Interface	TTL
A	1	...
E	4	...
B	2	...

Forwarding table: S2

MAC	Interface	TTL
A	4	...
E	2	...
B	4	...

① ^{src} A → ^{dst} D
broadcast

② ^{src} E → ^{dst} A

→ ③ B → C

→ ④ Switch S2 receives a frame from port '4' with dst 'B'